

Module 3: Protocols and Models

CCNA 1: Introduction to Networks v7.0 — Comprehensive Reviewer & Question Bank

Student Name: _____

Date: _____

Class / Section: _____

Score: _____ / 95

PART I: MULTIPLE CHOICE IDENTIFICATION

General Instructions: Read each statement carefully. Select the most accurate term, acronym, or concept that satisfies the blank. Write the letter of your choice on the blank before each number. (95 items)

1. In any network communication, the sending device where a message originates is known as the _____.
 (A) channel (media) (B) destination (receiver) (C) source (sender) (D) default gateway

2. In network communications, the receiving device that accepts and decodes the message is called the _____.
 (A) repeater (B) destination (receiver) (C) channel (media) (D) source (sender)

3. The physical medium or path providing the transmission pathway over which messages travel from sender to receiver is called the _____.
 (A) protocol suite (B) frame check sequence (C) channel (media) (D) data link layer

4. The three fundamental elements present in every network communication are the source, destination, and _____.
 (A) channel (media) (B) firewall (C) default gateway (D) router

5. The established rules and agreements that govern how devices communicate across a network are known as _____.
 (A) latencies (B) topologies (C) protocols (D) bandwidths

_____	6. A human or computer conversation cannot proceed effectively without an identified sender, receiver, common language, speed/timing, and confirmation or _____ requirements. (A) multiplexing (B) segmentation (C) acknowledgment (D) encryption
_____	7. The process of converting human-readable data or information into another acceptable electronic or optical form for transmission across a medium is called _____. (A) routing (B) decoding (C) de-encapsulation (D) encoding
_____	8. The process where a receiving device reverses the encoding process to convert signals back into readable information is called _____. (A) decoding (B) segmentation (C) encapsulation (D) multiplexing
_____	9. When messages are sent across a network medium, they are broken down and transmitted as patterns of electrical impulses, sound waves, or light representing _____. (A) frames (B) bits (C) packets (D) segments
_____	10. The timing mechanism that manages the rate of data transmission to prevent a fast sender from overwhelming a slow receiver is called _____. (A) response timeout (B) access method (C) flow control (D) carrier sensing
_____	11. The timing rule that dictates how long a transmitting device waits for an acknowledgment or reply before assuming the message was lost is called the _____. (A) propagation delay (B) flow control (C) response timeout (D) jitter buffer
_____	12. The rule that determines when a device is permitted to send a

	message across shared media to prevent transmission conflicts is the ____. (A) session timeout (B) flow control (C) modulation method (D) access method
_____	13. A condition where two or more devices simultaneously transmit data signals across a shared medium causing signals to corrupt each other is called a ____. (A) crossover (B) collision (C) broadcast storm (D) deadlock
_____	14. The message delivery method where information is transmitted from a single source host directly to a single destination host is called ____. (A) multicast (B) broadcast (C) unicast (D) anycast
_____	15. The message delivery option where a message is sent from one source to a selected group of destination hosts that subscribed to receive it is called ____. (A) anycast (B) multicast (C) broadcast (D) unicast
_____	16. The message delivery method in IPv4 where a packet is sent from a single sender to all hosts on the local network segment is called ____. (A) multicast (B) broadcast (C) unicast (D) anycast
_____	17. Unlike IPv4, the IPv6 protocol does not implement broadcast messaging; instead, it uses multicast and a special one-to-nearest delivery option called ____. (A) unicast only (B) geocast (C) simulcast (D) anycast
_____	18. In Cisco network documentation and topology diagrams, client computers, servers, and intermediate devices are frequently represented using a generic circular symbol called the ____.

(A) bus bar (B) bridge glyph (C) node icon (D) cloud token

- ____
19. Network protocols can be implemented in device hardware, device software, or ____.
- (A) firmware only (B) cloud storage only (C) both hardware and software (D) virtual machines only
-

- ____
20. The protocol category responsible for enabling two or more devices to exchange data packets over one or more interconnected networks is ____.
- (A) network communications (B) service discovery (C) routing (D) network security
-

- ____
21. The protocol category that provides data encryption, user authentication, and data integrity verification is ____.
- (A) service discovery (B) network security (C) routing (D) multiplexing
-

- ____
22. The protocol type that enables routers to exchange network path information, compare metrics, and select the best path to destination networks is ____.
- (A) error detection (B) flow control (C) service discovery (D) routing
-

- ____
23. The protocol category used for the automatic detection of devices, network services, and resources on a local network without manual configuration is ____.
- (A) service discovery (B) congestion control (C) routing (D) network security
-

- ____
24. The network protocol function that uniquely identifies the sending device and receiving device on a network is ____.
- (A) flow control (B) reliability (C) addressing (D) sequencing
-

- ____
25. The protocol function that guarantees
-

the delivery of data and provides retransmission if packets are lost during transit is ____.

- (A) reliability (B) carrier sense
(C) service discovery (D) best-effort delivery
-

26. The protocol function that numbers or uniquely labels each transmitted data piece so that they can be reassembled in the correct order is called ____.

- (A) multiplexing (B) sequencing
(C) addressing (D) error detection
-

27. The protocol function that determines whether transmitted bits became corrupted or altered during transmission across the physical media is ____.

- (A) access control (B) flow control
(C) error simulation (D) error detection
-

28. The protocol function that manages direct process-to-process communication between software applications running on different hosts is the ____.

- (A) routing table (B) channel arbitration
(C) flow control (D) application interface
-

29. The application layer protocol that governs how web browsers (clients) request web pages from web servers is ____.

- (A) SNMP (B) HTTP (Hypertext Transfer Protocol) (C) FTP (D) SMTP
-

30. In web traffic, the transport layer protocol that manages individual conversations, divides data into segments, and guarantees reliable delivery is ____.

- (A) TCP (Transmission Control Protocol) (B) UDP (C) IP (D) ICMP
-

31. The internet layer protocol that encapsulates segments into packets

and delivers them globally across intermediate routers from source to destination is ____.

- (A) TCP (B) Ethernet (C) IP
(Internet Protocol) (D) ARP
-

- ____
32. The network access layer protocol standard that delivers data frames from one network interface card (NIC) to another NIC on the same local Ethernet LAN is ____.

- (A) IP (B) TCP (C) HTTP (D)
Ethernet
-

- ____
33. A group of inter-related protocols that work cooperatively across different layers to perform network communication functions is called a ____.

- (A) protocol suite (B) domain
namespace (C) routing table (D)
broadcast domain
-

- ____
34. In a layered protocol suite, the lower layers are primarily responsible for moving data across the physical media and providing services to the ____.

- (A) upper layers (B) BIOS only
(C) adjacent routers only (D)
firmware chip only
-

- ____
35. The universal, open-standard protocol suite that powers the modern internet and enterprise networks is the ____ suite.

- (A) Novell NetWare (B) TCP/IP
(C) AppleTalk (D) Decnet
-

- ____
36. The TCP/IP protocol suite is maintained, updated, and standardized by the ____.

- (A) EIA (B) ITU-T (C) IEEE (D)
IETF (Internet Engineering Task
Force)
-

- ____
37. The historical proprietary protocol suite developed and released by Apple Inc. for Macintosh local networking in the 1980s was ____.

- (A) OSI Suite (B) AppleTalk (C)
DECnet (D) Novell NetWare
-

_____	38. The historical proprietary protocol suite that included IPX/SPX and was widely used in early enterprise local area networks was _____. (A) Novell NetWare (B) SNA (C) AppleTalk (D) Ethernet II
_____	39. A protocol specification that is published without proprietary vendor restrictions, freely available to the public, and usable by anyone is called an _____. (A) open standard (B) commercial license (C) closed proprietary system (D) trade secret
_____	40. A protocol suite that is endorsed by the networking industry and ratified by an official standards organization to guarantee multi-vendor interoperability is termed _____. (A) ad-hoc (B) vendor-locked (C) proprietary (D) standards-based
_____	41. The two most prevalent Network Access layer local area network (LAN) technologies in the TCP/IP model are Ethernet and _____. (A) WLAN (Wireless LAN) (B) Token Ring (C) ATM (D) FDDI
_____	42. Standards organizations that develop network protocols must be non-profit, open to the public, and _____. (A) vendor-neutral (B) privately commercial (C) vendor-specific (D) government-operated only
_____	43. Open networking standards encourage three major market benefits: interoperability, innovation, and _____. (A) monopoly (B) vendor lock-in (C) planned obsolescence (D) competition
_____	44. The non-profit organization dedicated to ensuring the open development, evolution, and ethical use of the Internet worldwide is the _____. (A) ISOC (Internet Society) (B)

IAB (C) ICANN (D) IANA

45. The committee responsible for overall architectural oversight, technical management, and appeal procedures for Internet standards is the _____.
(A) IETF (B) IEEE (C) IAB
(Internet Architecture Board) (D) IRTF

46. The working group organization that develops, tests, and maintains internet standards and TCP/IP protocols via RFC documents is the _____.
(A) IRTF (B) ITU-T (C) IETF
(Internet Engineering Task Force)
(D) ICANN

47. The organization that focuses on long-term theoretical research topics related to the Internet, TCP/IP, and future protocol architectures is the _____.
(A) IRTF (Internet Research Task Force) (B) ISOC (C) IETF (D) EIA

48. The international non-profit entity that coordinates global IP address space allocation, top-level domain (TLD) names, and protocol identifiers is _____.
(A) ICANN (Internet Corporation for Assigned Names and Numbers)
(B) ISOC (C) IETF (D) IANA

49. The operational department overseen by ICANN that directly manages the global allocation of IPv4/IPv6 address blocks, AS numbers, and root DNS zones is _____.
(A) IRTF (B) IETF (C) IAB (D) IANA (Internet Assigned Numbers Authority)

50. The global professional organization that created the 802 LAN/MAN standards, including 802.3 Ethernet and 802.11 Wi-Fi, is the _____.
(A) TIA (B) EIA (C) ITU-T (D) IEEE (Institute of Electrical and

51. The trade organization known for standardizing electrical wiring pinouts, serial connectors, and standard 19-inch equipment racks is the _____.

- (A) ISOC (B) IEEE (C) ICANN
(D) EIA (Electronic Industries Alliance)

52. The standards body responsible for developing communication standards in radio equipment, cellular towers, VoIP terminals, and satellite communications is the _____.

- (A) IAB (B) IANA (C) TIA
(Telecommunications Industry Association) (D) IRTF

53. The specialized United Nations agency sector responsible for international standards in video compression, IPTV, and broadband communications like DSL is the _____.

- (A) IETF (B) ITU-T
(Telecommunication Standardization Sector of ITU) (C) IEEE (D) ICANN

54. One primary advantage of using a layered networking model is that changes in technology or capabilities at one layer do not affect _____.

- (A) the speed of light (B) physical cable lengths (C) the destination host IP address (D) other layers above and below

55. A layered model fosters healthy vendor competition because hardware and software from different vendors can _____.

- (A) bypass standard security policies (B) avoid electrical grounding (C) interoperate (work together) (D) be purchased from a single supplier

56. How many distinct architectural layers are defined in the theoretical OSI reference model?

- (A) 4 (B) 8 (C) 5 (D) 7
-

_____	57. How many architectural layers are defined in the practical TCP/IP reference model? (A) 4 (B) 7 (C) 6 (D) 5
_____	58. In the OSI reference model, Layer 7 is the _____ layer. (A) Session (B) Application (C) Transport (D) Presentation
_____	59. In the OSI model, the layer responsible for data formatting, data compression, and data encryption/decryption between applications is the _____ layer. (A) Application (Layer 7) (B) Presentation (Layer 6) (C) Session (Layer 5) (D) Transport (Layer 4)
_____	60. The OSI model layer that establishes, manages, synchronizes, and terminates communication dialogues between cooperating applications is the _____ layer. (A) Session (Layer 5) (B) Network (Layer 3) (C) Presentation (Layer 6) (D) Transport (Layer 4)
_____	61. The OSI layer that segments, transfers, and reassembles data for individual communications between end hosts is the _____ layer. (A) Network (Layer 3) (B) Session (Layer 5) (C) Transport (Layer 4) (D) Data Link (Layer 2)
_____	62. The OSI layer responsible for logical addressing, path determination, and forwarding packets across intermediate networks is the _____ layer. (A) Network (Layer 3) (B) Data Link (Layer 2) (C) Transport (Layer 4) (D) Physical (Layer 1)
_____	63. The OSI layer responsible for exchanging data frames across a common physical medium and managing physical MAC addressing is the _____ layer. (A) Physical (Layer 1) (B) Session

(Layer 5) **(C)** Data Link (Layer 2)
(D) Network (Layer 3)

- _____
- 64.** The OSI model layer responsible for the physical connectors, cabling specifications, and transmitting raw bit signals across transmission media is the _____ layer.
- (A)** Data Link (Layer 2) **(B)** Network (Layer 3) **(C)** Transport (Layer 4) **(D)** Physical (Layer 1)
-

- _____
- 65.** In the TCP/IP model, the single top layer that combines the functions of the OSI Application, Presentation, and Session layers is the _____ layer.
- (A)** Network Access **(B)** Application **(C)** Transport **(D)** Internet
-

- _____
- 66.** The TCP/IP layer that corresponds directly to OSI Layer 3 and determines the best path through interconnected networks is the _____ layer.
- (A)** Internet **(B)** Application **(C)** Network Access **(D)** Transport
-

- _____
- 67.** The TCP/IP layer that controls the physical hardware devices and media and corresponds to OSI Layers 1 and 2 is the _____ layer.
- (A)** Internet **(B)** Network Access **(C)** Physical Only **(D)** Transport
-

- _____
- 68.** The networking process of dividing a large stream of application data into smaller, manageable chunks before transmission is known as _____.
- (A)** encapsulation **(B)** routing **(C)** segmenting (segmentation) **(D)** multiplexing
-

- _____
- 69.** The technique of taking multiple streams of segmented data from different conversations and interleaving them together over a single network channel is called _____.
- (A)** de-encapsulation **(B)** modulation **(C)** broadcasting **(D)** multiplexing
-

_____	70. A major benefit of segmenting data messages is that if a piece of data fails to reach its destination, only the lost _____ needs to be retransmitted. (A) segment (B) entire file (C) entire database (D) operating system
_____	71. The formal term used to describe the generic package of data and protocol control information created at each layer of a networking model is a _____. (A) MAC table entry (B) PDU (Protocol Data Unit) (C) SYN packet (D) MTU (Maximum Transmission Unit)
_____	72. In the TCP/IP protocol suite, what is the specific name of the PDU at the Transport layer? (A) Packet (B) Frame (C) Bits (D) Segment
_____	73. In the TCP/IP protocol suite, what is the specific name of the PDU at the Internet (Network) layer? (A) Frame (B) Packet (C) Bits (D) Segment
_____	74. In the TCP/IP protocol suite, what is the specific name of the PDU at the Data Link layer? (A) Frame (B) Bits (C) Packet (D) Segment
_____	75. At the Physical layer, what is the name of the PDU transmitted across the physical media as electrical, optical, or radio signals? (A) Bits (B) Segments (C) Packets (D) Frames
_____	76. What is the correct top-down sequence of Protocol Data Units (PDUs) as data moves down the TCP/IP stack during encapsulation? (A) Bits → Frame → Packet → Segment → Data (B) Data → Segment → Packet → Frame → Bits (C) Frame → Packet → Segment → Data → Bits (D) Data

→ Packet → Segment → Frame → Bits

- _____
- 77.** The top-down process where each layer adds its own protocol control header (and trailer) to the data received from the layer above is called _____.
(A) demultiplexing (B) encapsulation (C) decryption (D) de-encapsulation
-

- _____
- 78.** The bottom-up process where a receiving host strips off protocol headers at each successive layer to deliver the original data to the application is called _____.
(A) fragmentation (B) de-encapsulation (C) compression (D) encapsulation
-

- _____
- 79.** In the encapsulation process, what transport layer protocol is specifically responsible for numbering and sequencing data segments?
(A) IP (B) TCP (C) UDP (D) ICMP
-

- _____
- 80.** The logical addressing information contained in the Layer 3 IP packet header is responsible for delivering the packet from its original source to its _____.
(A) final destination (B) next-hop router only (C) local switch only (D) default gateway MAC
-

- _____
- 81.** The physical addressing information contained in the Layer 2 Data Link frame header is responsible for delivering data from one _____ to another on the same link.
(A) NIC (Network Interface Card) (B) domain name server (C) autonomous system (D) web server port
-

- _____
- 82.** An IPv4 address consists of two functional sections: the host portion and the _____ portion.
(A) MAC (B) port (C) prefix only (D) network
-

_____	83. In an IPv6 address, the left-most portion that identifies the network group is called the ____. (A) host portion (B) interface ID (C) prefix (D) socket
_____	84. In an IPv6 address, the right-most 64 bits that uniquely identify a specific host device on the local network are called the ____. (A) Subnet Mask (B) MAC Tag (C) Interface ID (D) Prefix
_____	85. When two communicating devices reside on the same local IP network, their IP addresses share the exact same ____ portion. (A) network (B) host (C) interface ID (D) port number
_____	86. When sending data to a host located on the exact same Ethernet network, what address is placed in the Destination MAC field of the frame? (A) The broadcast MAC address FF-FF-FF-FF-FF-FF (B) The MAC address of the default gateway router (C) The source device's own MAC address (D) The MAC address of the destination device's NIC
_____	87. Physical MAC addresses are permanently burned or embedded into the hardware of a device's ____. (A) NIC (Network Interface Card) (B) operating system registry (C) hard drive (D) RAM cache
_____	88. When a host determines that the destination IP address belongs to a remote network, it forwards the packet to the local router interface known as the ____. (A) DNS server (B) default gateway (C) broadcast address (D) loopback address
_____	89. If end devices on a local area network are not configured with an accurate default gateway IP address, their outbound network traffic will be

confined strictly to the ____.

- (A) local LAN (B) remote WAN
(C) Internet (D) cloud provider
-

- ____
90. When a source host (PC1) sends a packet to a remote web server across multiple routers, the destination MAC address in the first transmitted frame will be the MAC address of the ____.
- (A) second router's exit interface
(B) default gateway (local router interface) (C) local host's own loopback (D) remote web server NIC
-

- ____
91. As an IP packet travels across multiple router hops toward a remote destination, the source and destination IP addresses in the packet header remain ____.
- (A) swapped at each hop (B) unchanged (C) incremented at each hop (D) replaced by MAC addresses
-

- ____
92. Unlike Layer 3 IP addresses which remain constant end-to-end, Layer 2 MAC addresses are stripped and replaced at every router ____.
- (A) hop (link) (B) TCP handshake
(C) DNS query (D) session restart
-

- ____
93. When the final router on the path receives a packet destined for a host on its directly connected LAN, it encapsulates the packet in a frame whose destination MAC is the ____.
- (A) first router's entry interface (B) switch management interface (C) final destination host NIC (D) Internet Service Provider gateway
-

- ____
94. In network data access, which layer provides local physical addressing that changes from hop to hop?
- (A) Application Layer (Layer 7) (B) Network Layer (Layer 3) (C) Transport Layer (Layer 4) (D) Data Link Layer (Layer 2)
-

- ____
95. In network data access, which layer provides global logical addressing
-

that remains constant from the original source to the final destination?

- (A) Data Link Layer (Layer 2) (B)
 - Physical Layer (Layer 1) (C)
 - Session Layer (Layer 5) (D)
 - Network Layer (Layer 3)
-

PART II: COMPLETE ANSWER KEY & TECHNICAL RATIONALES

Authoritative Cisco ITN v7.0 Reference Key — Module 3: Protocols and Models

Item #	Key	Correct Answer / Term	Topic & Explanation
1	C.)	source (sender)	[3.1 The Rules - Fundamentals] All communication begins with a source (sender) that creates and sends the message.
2	B.)	destination (receiver)	[3.1 The Rules - Fundamentals] The destination (receiver) is the target device that receives and interprets the transmitted message.
3	C.)	channel (media)	[3.1 The Rules - Fundamentals] The channel consists of the media (copper wires, optical fibers, or wireless radio frequencies) that provides the physical path for communication.
4	A.)	channel (media)	[3.1 The Rules - Fundamentals] Every communication requires three elements: a source (sender), a destination (receiver), and a channel (media).
5	C.)	protocols	[3.1 The Rules - Protocols] Protocols are the formal sets of rules and standards that govern data communications.
6	C.)	acknowledgment	[3.1 The Rules - Rule Establishment] Effective communication requires rule establishment including acknowledgment (confirmation that messages were received).
7	D.)	encoding	[3.1 The Rules - Message Encoding] Encoding converts data into an acceptable form for transmission over the physical channel.
8	A.)	decoding	[3.1 The Rules - Message Encoding] Decoding reverses encoding at the destination host to reconstruct the original message.
9	B.)	bits	[3.1 The Rules - Message Size] Messages on the network are ultimately

			converted into bits and encoded into physical signals (light, electricity, or radio waves).
10	C.)	flow control	[3.1 The Rules - Message Timing] Flow control manages transmission rate, defining how much information can be sent and at what speed.
11	C.)	response timeout	[3.1 The Rules - Message Timing] Response timeout manages how long a sending device waits for a reply before retransmitting or aborting.
12	D.)	access method	[3.1 The Rules - Message Timing] The access method determines when a device can transmit onto shared network media.
13	B.)	collision	[3.1 The Rules - Message Timing] A collision occurs when multiple devices transmit traffic at the exact same moment on a shared channel, corrupting both messages.
14	C.)	unicast	[3.1 The Rules - Delivery Options] Unicast refers to one-to-one communication between a single sender and single receiver.
15	B.)	multicast	[3.1 The Rules - Delivery Options] Multicast is one-to-many communication, delivering to a specific subset of hosts.
16	B.)	broadcast	[3.1 The Rules - Delivery Options] Broadcast is one-to-all communication on a local network segment.
17	D.)	anycast	[3.1 The Rules - Delivery Options] IPv6 eliminated traditional broadcast in favor of multicast and anycast (one-to-nearest).
18	C.)	node icon	[3.1 The Rules - Node Icon] Network diagrams frequently use a circular node icon to represent any network-connected device.
19	C.)	both hardware and software	[3.2 Protocols - Overview] Protocols can be implemented

			in software (like OS drivers and applications), hardware (like NIC chipsets), or both.
20	A.)	network communications	[3.2 Protocols - Protocol Types] Network communications protocols (e.g., IP, TCP, HTTP) enable devices to communicate across networks.
21	B.)	network security	[3.2 Protocols - Protocol Types] Network security protocols (like SSH, TLS/SSL, IPsec) provide confidentiality, integrity, and authentication.
22	D.)	routing	[3.2 Protocols - Protocol Types] Routing protocols (e.g., OSPF, EIGRP, BGP) discover routes and calculate optimal forwarding paths.
23	A.)	service discovery	[3.2 Protocols - Protocol Types] Service discovery protocols (e.g., DNS-SD, DHCP, mDNS) automatically identify available network services.
24	C.)	addressing	[3.2 Protocols - Functions] Addressing provides source and destination identification (such as MAC addresses and IP addresses).
25	A.)	reliability	[3.2 Protocols - Functions] Reliability services (such as TCP acknowledgments and retransmissions) guarantee that sent data arrives intact.
26	B.)	sequencing	[3.2 Protocols - Functions] Sequencing assigns sequence numbers to segments so the receiver can reassemble them in the original sequence.
27	D.)	error detection	[3.2 Protocols - Functions] Error detection (such as checksums or Frame Check Sequence / CRC) detects damaged bits during transmission.
28	D.)	application interface	[3.2 Protocols - Functions] The application interface function enables inter-process communication between host applications (e.g., port numbers and APIs).

29	B.)	HTTP (Hypertext Transfer Protocol)	[3.2 Protocols - Interaction] HTTP defines the format and interactions between web clients and web servers.
30	A.)	TCP (Transmission Control Protocol)	[3.2 Protocols - Interaction] TCP provides connection-oriented, reliable transmission, flow control, and segment sequencing.
31	C.)	IP (Internet Protocol)	[3.2 Protocols - Interaction] IP handles logical addressing and delivers packets across interconnected networks.
32	D.)	Ethernet	[3.2 Protocols - Interaction] Ethernet governs local physical transmission and frame delivery between NICs on a LAN.
33	A.)	protocol suite	[3.3 Protocol Suites - Overview] A protocol suite is an organized collection of complementary protocols that solve communication problems together.
34	A.)	upper layers	[3.3 Protocol Suites - Layer Architecture] Lower layers handle physical transmission and data delivery, offering services to the upper layers.
35	B.)	TCP/IP	[3.3 Protocol Suites - Evolution] The TCP/IP suite is the dominant protocol suite used across the global Internet and local networks.
36	D.)	IETF (Internet Engineering Task Force)	[3.3 Protocol Suites - Evolution] The IETF manages and standardizes the protocols of the TCP/IP suite through RFC documents.
37	B.)	AppleTalk	[3.3 Protocol Suites - Evolution] AppleTalk was Apple's proprietary suite before adopting the standard TCP/IP model.
38	A.)	Novell NetWare	[3.3 Protocol Suites - Evolution] Novell NetWare used IPX/SPX as its proprietary protocol stack before modern TCP/IP dominance.
39	A.)	open standard	[3.3 Protocol Suites - Open Standards] Open standards

			ensure vendor neutrality and allow any company or developer to implement the technology.
40	D.)	standards-based	[3.3 Protocol Suites - Standards] Standards-based protocols have been approved by standards organizations to ensure interoperability between diverse manufacturers.
41	A.)	WLAN (Wireless LAN)	[3.3 Protocol Suites - TCP/IP Examples] Ethernet (wired) and WLAN (wireless, IEEE 802.11) are the two primary network access LAN technologies.
42	A.)	vendor-neutral	[3.4 Standards Organizations - Open Standards] Standards organizations remain vendor-neutral so no single corporation controls fundamental network standards.
43	D.)	competition	[3.4 Standards Organizations - Benefits] Open standards encourage interoperability, market competition, and rapid innovation.
44	A.)	ISOC (Internet Society)	[3.4 Standards Organizations - Internet Bodies] The Internet Society (ISOC) promotes open development and global evolution of the Internet.
45	C.)	IAB (Internet Architecture Board)	[3.4 Standards Organizations - Internet Bodies] The IAB manages internet architecture, advises ISOC, and supervises IETF standards processes.
46	C.)	IETF (Internet Engineering Task Force)	[3.4 Standards Organizations - Internet Bodies] The IETF produces RFC (Request for Comments) documents that define Internet and TCP/IP standards.
47	A.)	IRTF (Internet Research Task Force)	[3.4 Standards Organizations - Internet Bodies] The IRTF focuses on long-term research groups,

			whereas the IETF focuses on short-to-medium term engineering.
48	A.)	ICANN (Internet Corporation for Assigned Names and Numbers)	[3.4 Standards Organizations - IP Coordination] ICANN coordinates IP address allocation, DNS domain management, and protocol parameters globally.
49	D.)	IANA (Internet Assigned Numbers Authority)	[3.4 Standards Organizations - IP Coordination] IANA performs the operational responsibility for assigning IP addresses, protocol numbers, and port assignments.
50	D.)	IEEE (Institute of Electrical and Electronics Engineers)	[3.4 Standards Organizations - Electronics Standards] IEEE defines fundamental electrical and computer engineering standards such as 802.3 (Ethernet) and 802.11 (WLAN).
51	D.)	EIA (Electronic Industries Alliance)	[3.4 Standards Organizations - Electronics Standards] EIA developed physical standards for electrical wiring, connectors, and equipment racks.
52	C.)	TIA (Telecommunications Industry Association)	[3.4 Standards Organizations - Electronics Standards] TIA standardizes telecommunications equipment, cabling (such as TIA/EIA-568), and radio frequencies.
53	B.)	ITU-T (Telecommunication Standardization Sector of ITU)	[3.4 Standards Organizations - Electronics Standards] ITU-T defines international telecommunications standards including video codecs (H.264), DSL, and optical transport.
54	D.)	other layers above and below	[3.5 Reference Models - Layered Benefits] Layer independence ensures changes or upgrades in one layer do not require redesigning adjacent layers.
55	C.)	interoperate (work	[3.5 Reference Models -

		together)	Layered Benefits] Standardized layer interfaces allow products from different manufacturers to interoperate seamlessly.
56	D.)	7	[3.5 Reference Models - OSI Model] The OSI model consists of 7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.
57	A.)	4	[3.5 Reference Models - TCP/IP Model] The TCP/IP model consists of 4 layers: Network Access, Internet, Transport, and Application.
58	B.)	Application	[3.5 Reference Models - OSI Model] Layer 7 of the OSI model is the Application layer, providing services directly to user network software.
59	B.)	Presentation (Layer 6)	[3.5 Reference Models - OSI Model] The Presentation layer ensures common data representation, data compression, and encryption.
60	A.)	Session (Layer 5)	[3.5 Reference Models - OSI Model] The Session layer manages dialogues and data exchange sessions between applications.
61	C.)	Transport (Layer 4)	[3.5 Reference Models - OSI Model] Layer 4 (Transport) defines services to segment, transfer, and reassemble conversation data.
62	A.)	Network (Layer 3)	[3.5 Reference Models - OSI Model] Layer 3 (Network) handles logical addressing (IP) and exchanges individual pieces of data across networks.
63	C.)	Data Link (Layer 2)	[3.5 Reference Models - OSI Model] Layer 2 (Data Link) describes methods for exchanging data frames over common physical media.
64	D.)	Physical (Layer 1)	[3.5 Reference Models - OSI Model] Layer 1 (Physical) specifies electrical, mechanical, and optical means to transmit raw bit

			streams.
65	B.)	Application	[3.5 Reference Models - TCP/IP Model] The TCP/IP Application layer represents data to the user and handles encoding, presentation, and dialog control.
66	A.)	Internet	[3.5 Reference Models - TCP/IP Model] The TCP/IP Internet layer maps to OSI Layer 3 (Network) and routes packets using logical IP addressing.
67	B.)	Network Access	[3.5 Reference Models - TCP/IP Model] The Network Access layer in the TCP/IP model encompasses both the Data Link and Physical layers of the OSI model.
68	C.)	segmenting (segmentation)	[3.6 Data Encapsulation - Segmenting] Segmenting breaks large messages into smaller units so they can travel across the network efficiently.
69	D.)	multiplexing	[3.6 Data Encapsulation - Multiplexing] Multiplexing interleaves multiple communication streams across shared transmission channels.
70	A.)	segment	[3.6 Data Encapsulation - Benefits] Segmentation increases efficiency because only failed segments must be retransmitted rather than the entire message.
71	B.)	PDU (Protocol Data Unit)	[3.6 Data Encapsulation - PDUs] A Protocol Data Unit (PDU) is the specific form that data takes at any given layer of the protocol stack.
72	D.)	Segment	[3.6 Data Encapsulation - PDUs] At the Transport layer, the PDU is called a Segment (or Datagram for UDP).
73	B.)	Packet	[3.6 Data Encapsulation - PDUs] At the Network/Internet layer, the PDU encapsulated with IP addressing is called a Packet.
74	A.)	Frame	[3.6 Data Encapsulation -

			PDUs] At the Data Link layer, data encapsulated with header and trailer information is called a Frame.
75	A.)	Bits	[3.6 Data Encapsulation - PDUs] At the Physical layer, the PDU consists of raw Bits transmitted over the physical medium.
76	B.)	Data → Segment → Packet → Frame → Bits	[3.6 Data Encapsulation - PDU Order] Data moves top-down: Data (Application) → Segment (Transport) → Packet (Internet) → Frame (Data Link) → Bits (Physical).
77	B.)	encapsulation	[3.6 Data Encapsulation - Process] Encapsulation adds layer-specific protocol headers as data moves down from the application layer to the physical media.
78	B.)	de-encapsulation	[3.6 Data Encapsulation - Process] De-encapsulation occurs at the receiving host, stripping each header as data moves up the protocol stack.
79	B.)	TCP	[3.6 Data Encapsulation - Sequencing] TCP tracks and orders transmitted chunks using sequence numbers to guarantee accurate reassembly.
80	A.)	final destination	[3.7 Data Access - Layer 3 Addressing] Layer 3 IP addresses deliver the packet from the original sending host to the ultimate final destination host across all networks.
81	A.)	NIC (Network Interface Card)	[3.7 Data Access - Layer 2 Addressing] Layer 2 MAC addresses govern delivery from one NIC to the next NIC on the same local network segment.
82	D.)	network	[3.7 Data Access - IP Structure] An IP address is hierarchically divided into a network portion (identifying the subnet) and a host portion (identifying the specific device).
83	C.)	prefix	[3.7 Data Access - IP

			Structure] In IPv6, the network part of the address is called the prefix, while the host part is the Interface ID.
84	C.)	Interface ID	[3.7 Data Access - IP Structure] The Interface ID identifies the individual host interface within an IPv6 network prefix.
85	A.)	network	[3.7 Data Access - Same Network] Devices on the same network or subnet share identical network portions in their IP addresses.
86	D.)	The MAC address of the destination device's NIC	[3.7 Data Access - Same Network] When the destination host is on the same local network, the destination MAC address is that host's physical NIC MAC.
87	A.)	NIC (Network Interface Card)	[3.7 Data Access - MAC Addressing] MAC addresses are hardware-level addresses burned into the ROM of a network interface card (NIC).
88	B.)	default gateway	[3.7 Data Access - Remote Network] The default gateway is the router interface on the local subnet that acts as the exit door to remote networks.
89	A.)	local LAN	[3.7 Data Access - Default Gateway] Without a configured default gateway, hosts can only communicate with other devices on their local LAN.
90	B.)	default gateway (local router interface)	[3.7 Data Access - Remote Frame Addressing] For remote destinations, the source host addresses the frame's destination MAC to its local default gateway router interface.
91	B.)	unchanged	[3.7 Data Access - Address Persistence] The original source IP and ultimate destination IP addresses do not change as the packet moves through intermediate routers.
92	A.)	hop (link)	[3.7 Data Access - Address Modification] At each router

			hop, the old Layer 2 frame is stripped off and a new frame with new source and destination MAC addresses is generated for the next link.
93	C.)	final destination host NIC	[3.7 Data Access - Final Delivery] On the final local subnet, the router frames the packet with the destination host's actual MAC address for final delivery.
94	D.)	Data Link Layer (Layer 2)	[3.7 Data Access - Summary] The Data Link layer provides local hop-to-hop physical addressing (MAC), whereas the Network layer provides global end-to-end logical addressing (IP).
95	D.)	Network Layer (Layer 3)	[3.7 Data Access - Summary] The Network layer (Layer 3) provides end-to-end logical addressing that does not change across routers.